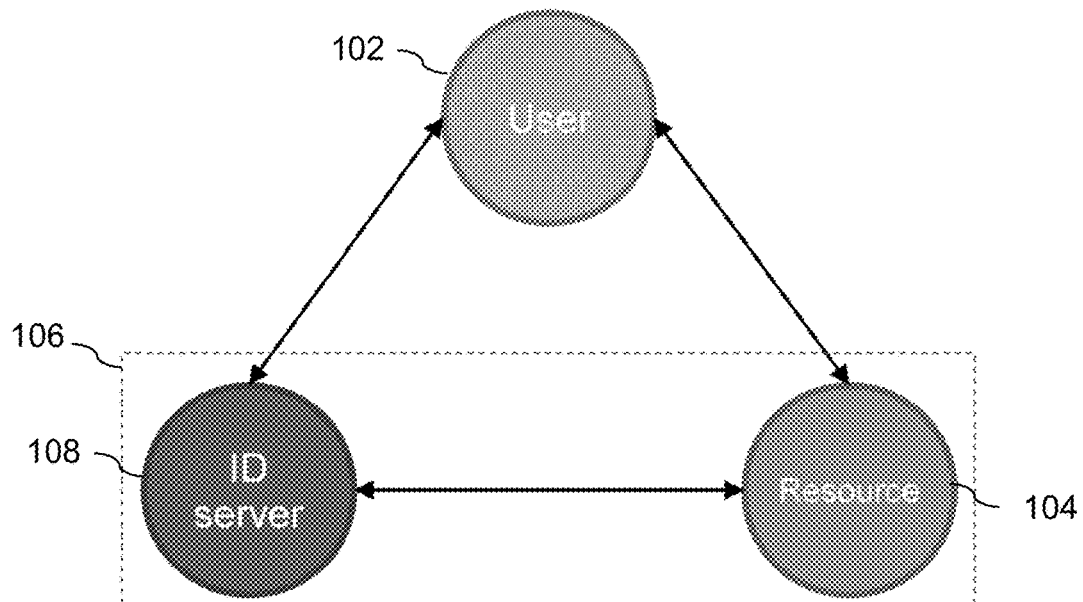


(19) **United States**(12) **Patent Application Publication****Wu et al.**(10) **Pub. No.: US 2020/0137064 A1**(43) **Pub. Date: Apr. 30, 2020**(54) **DECENTRALIZED IDENTITY
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Kun Wang, Beijing (CN)(21) Appl. No.: **16/173,446**(22) Filed: **Oct. 29, 2018****Publication Classification**(51) **Int. Cl.**
H04L 29/06 (2006.01)
H04L 9/00 (2006.01)**G06F 17/30** (2006.01)**H04L 9/06** (2006.01)(52) **U.S. Cl.**CPC **H04L 63/102** (2013.01); **H04L 63/0823**
(2013.01); **H04L 2209/38** (2013.01); **G06F**
17/30194 (2013.01); **H04L 9/0637** (2013.01);
H04L 9/006 (2013.01)

(57)

ABSTRACT

A given node associated with a plurality of nodes registers a decentralized identity for the given node on a decentralized identity blockchain. The registered decentralized identity is controlled by the given node and defined by an identity record stored on the decentralized identity blockchain. The registered decentralized identity for the given node is used to access one or more resources of a given decentralized application.

100

100

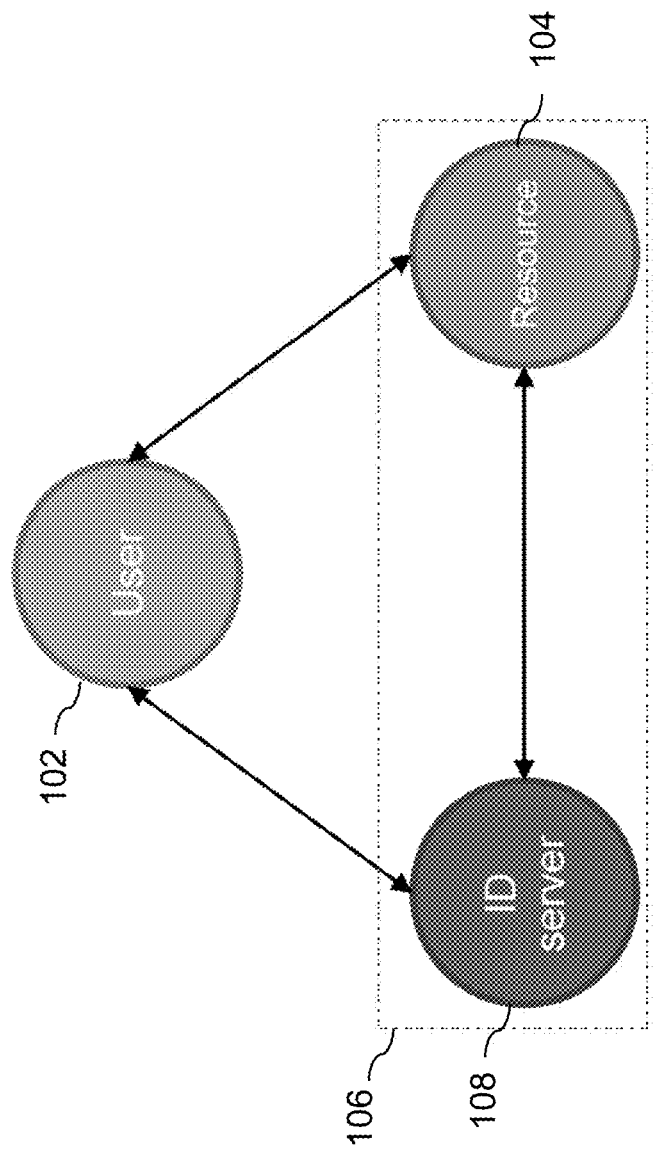


FIG. 1

200

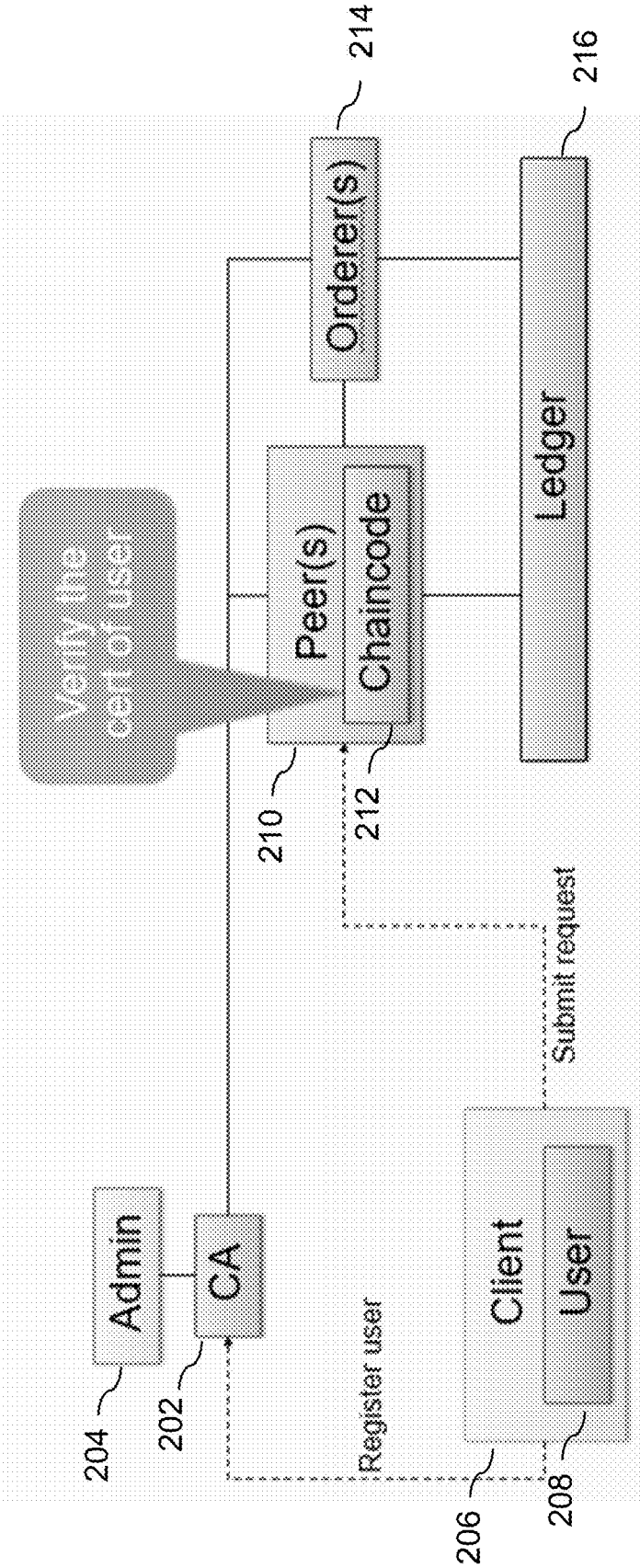


FIG. 2

300

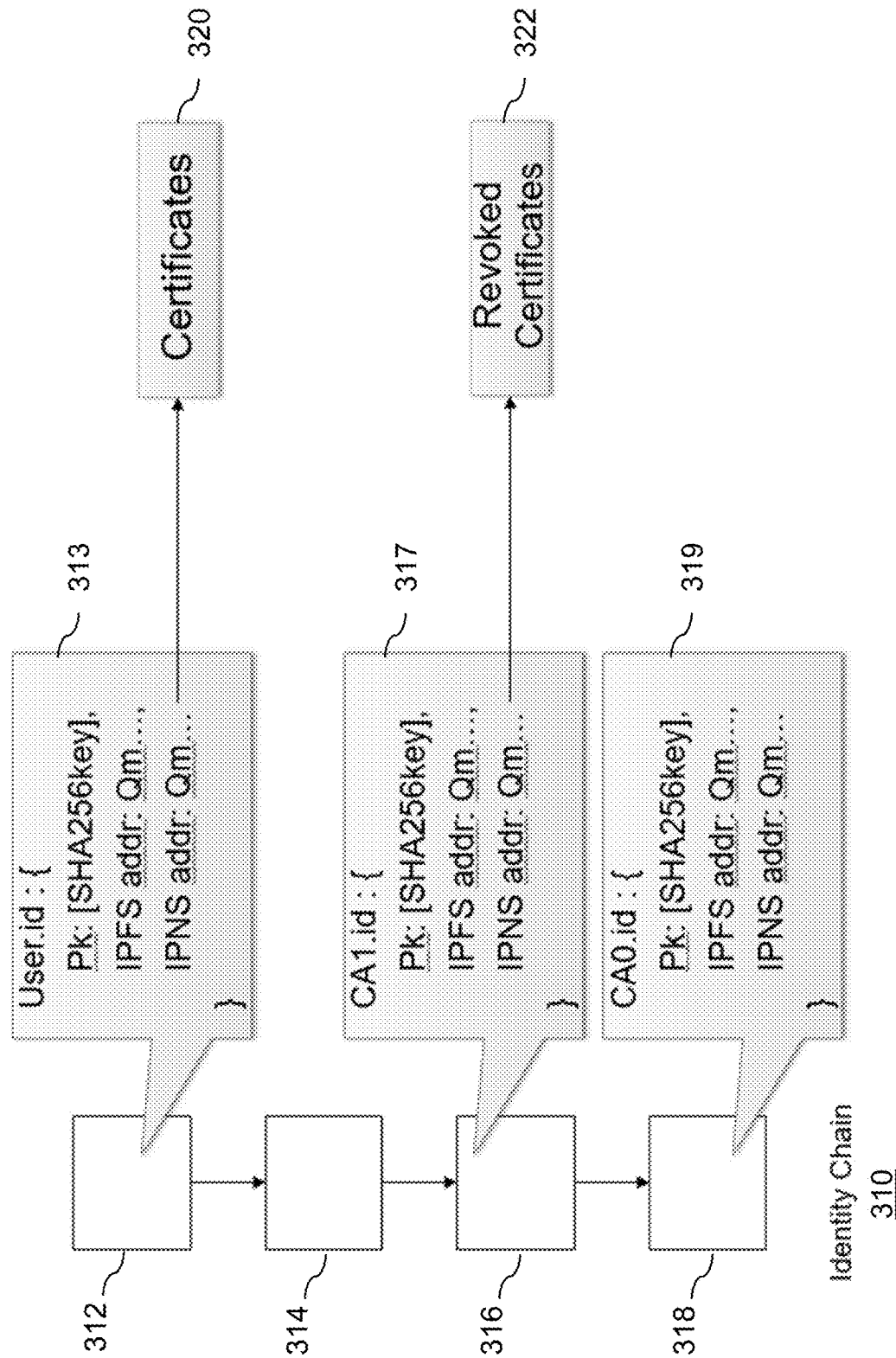


FIG. 3

400

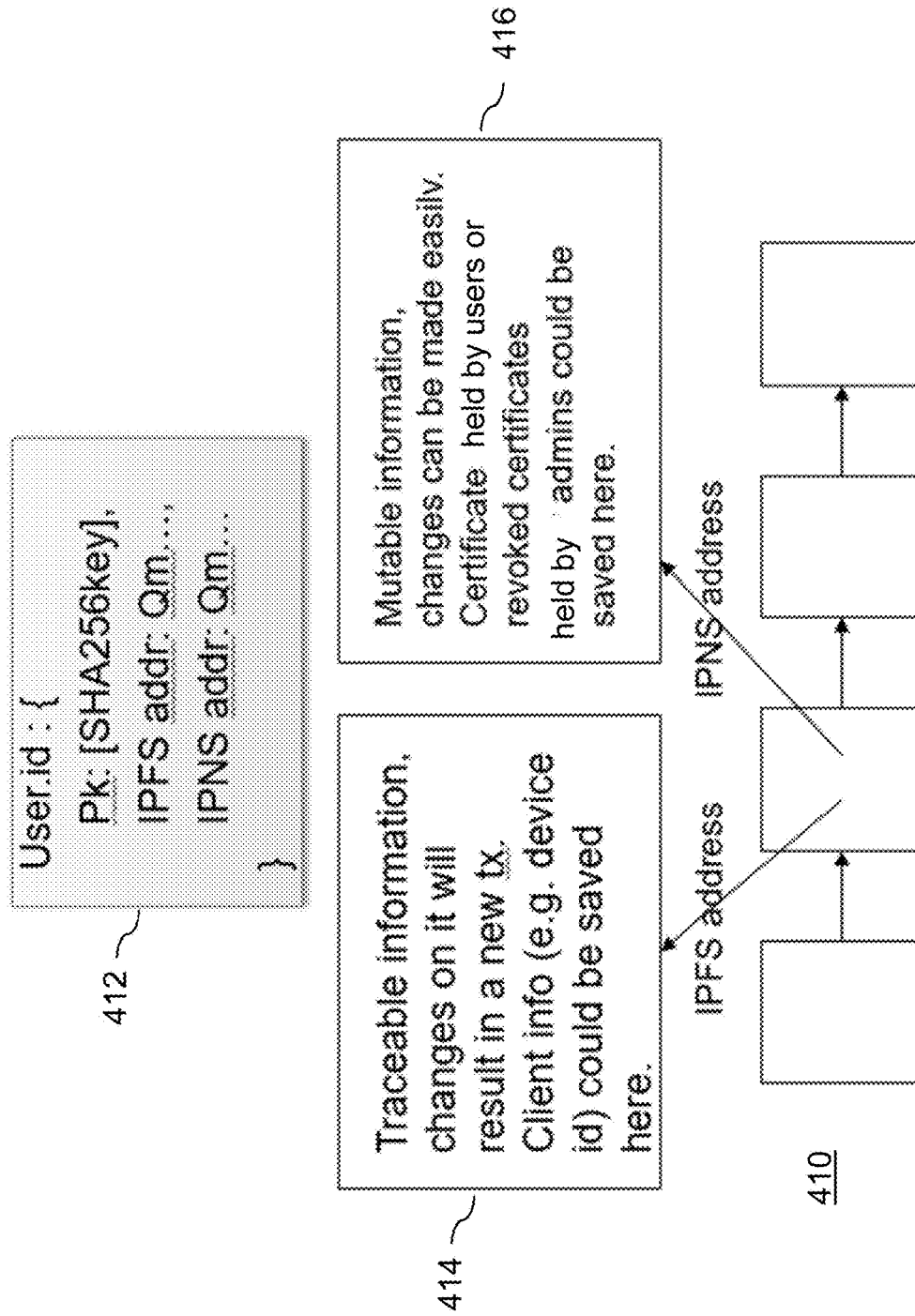


FIG. 4

500

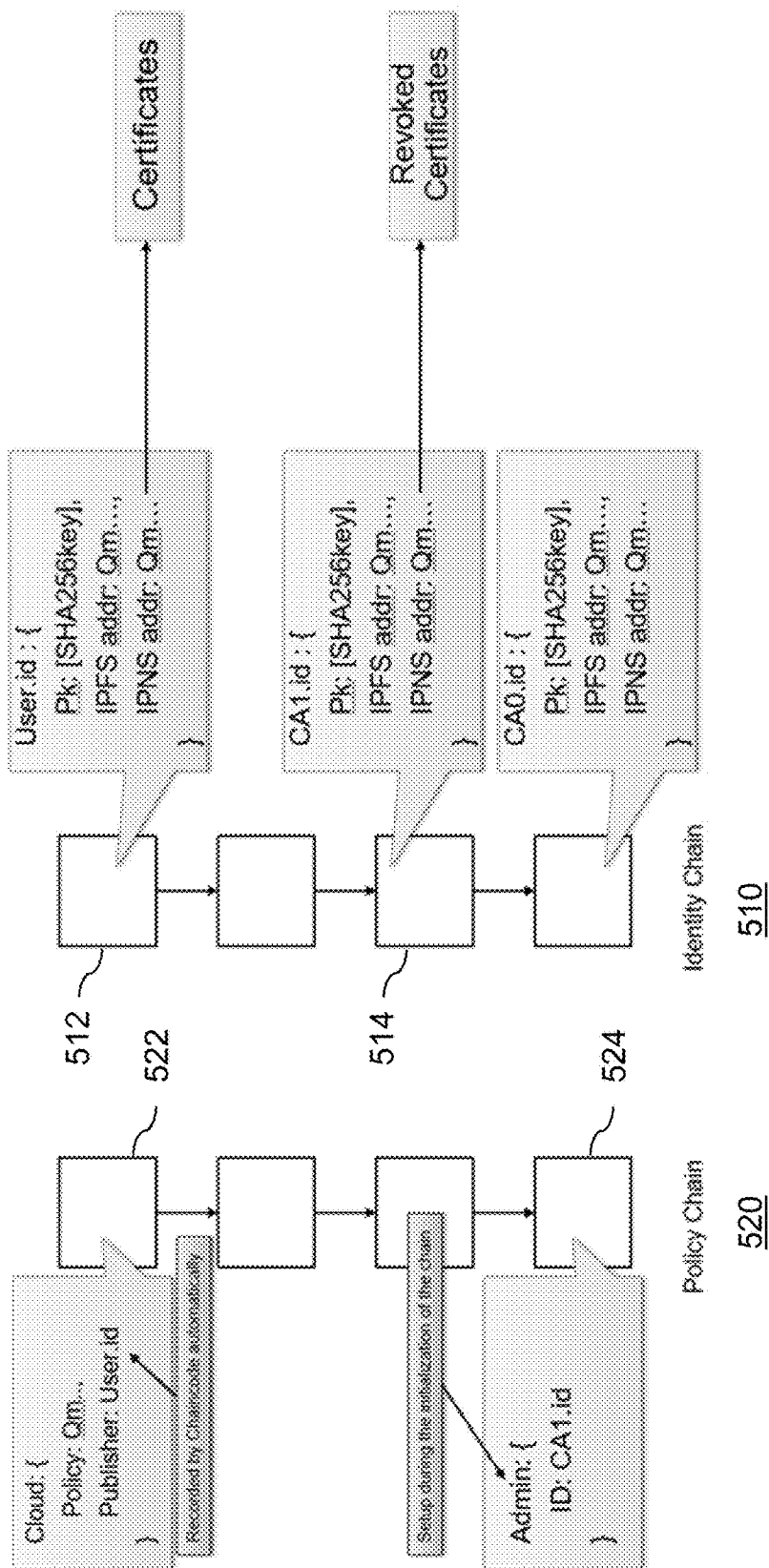


FIG. 5

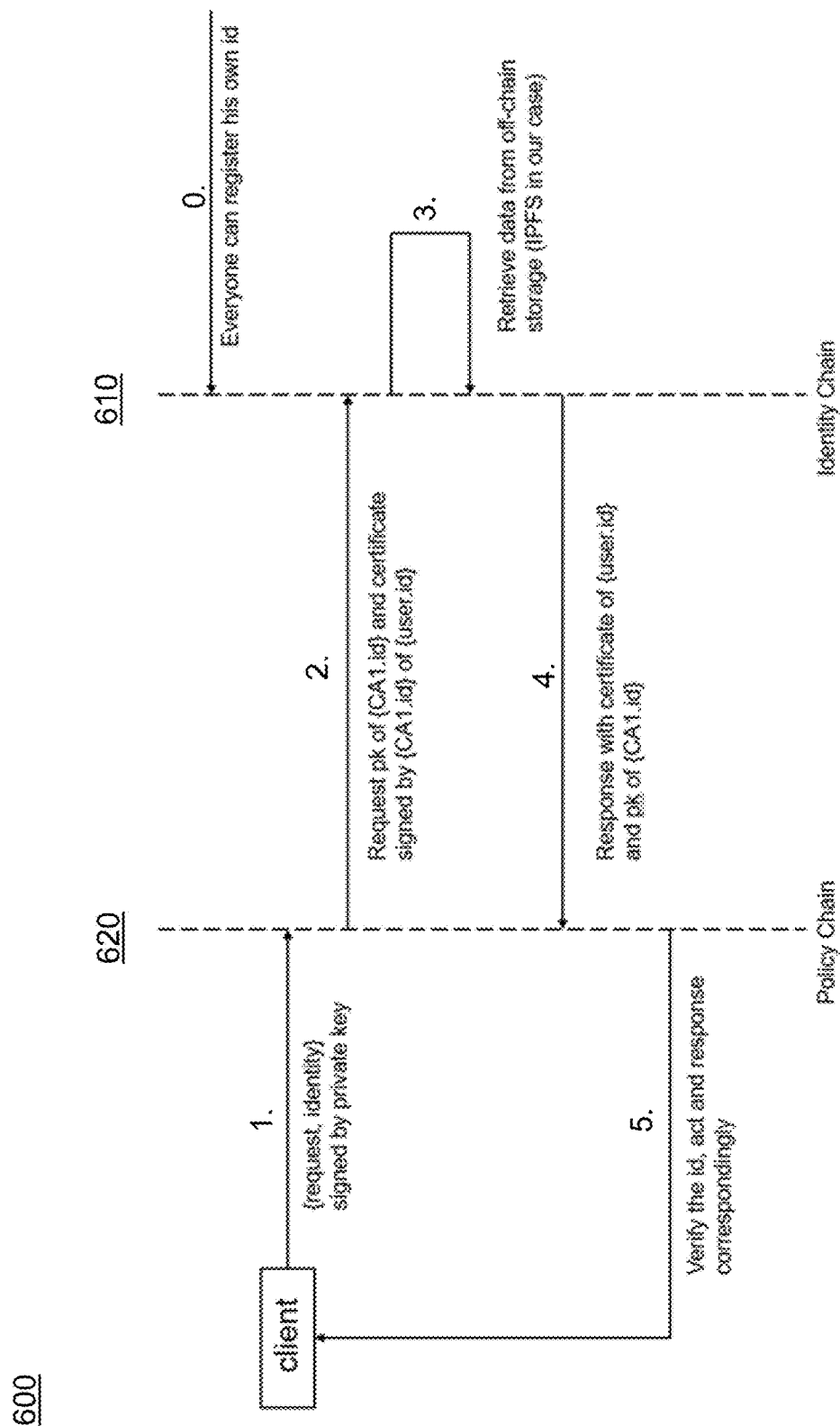


FIG. 6

700

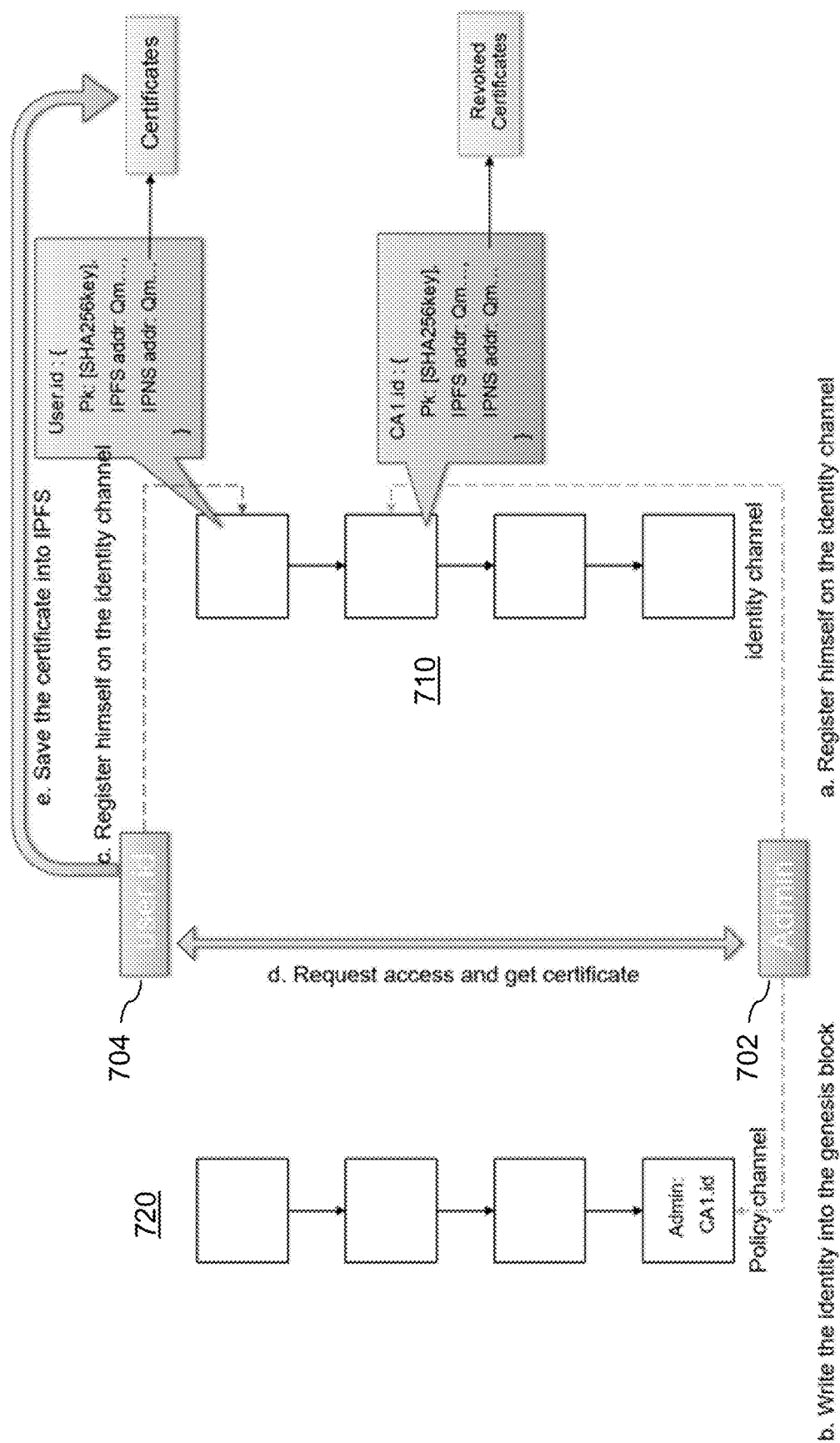


FIG. 7

800

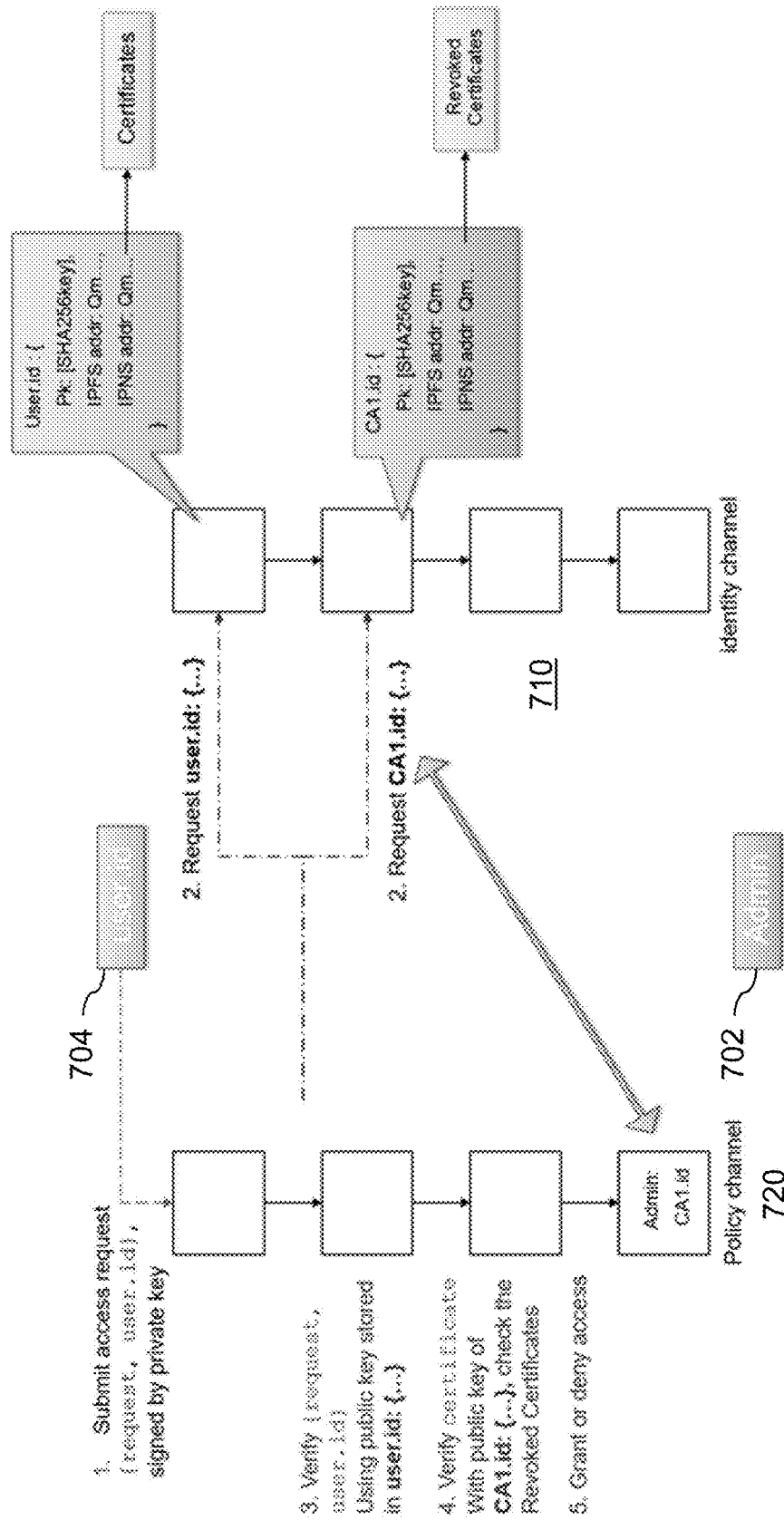


FIG. 8

900

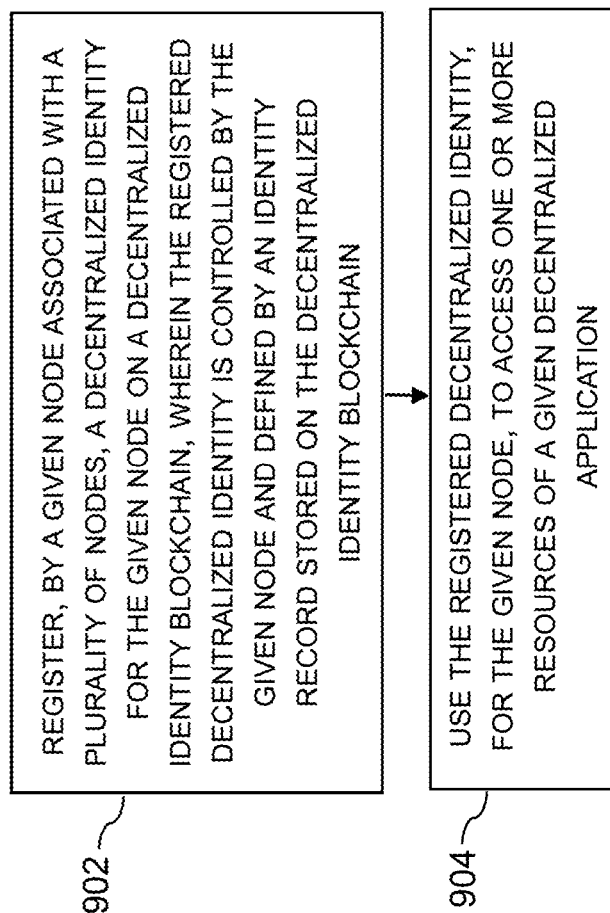


FIG. 9

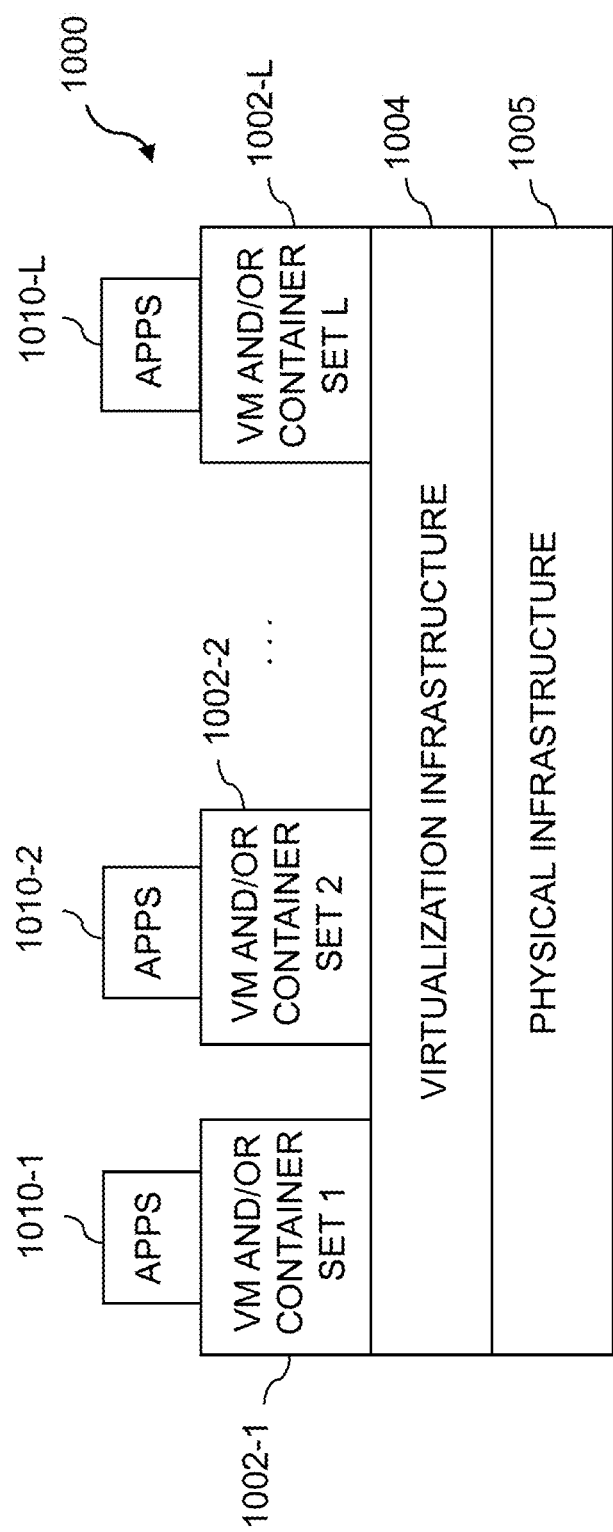


FIG. 10

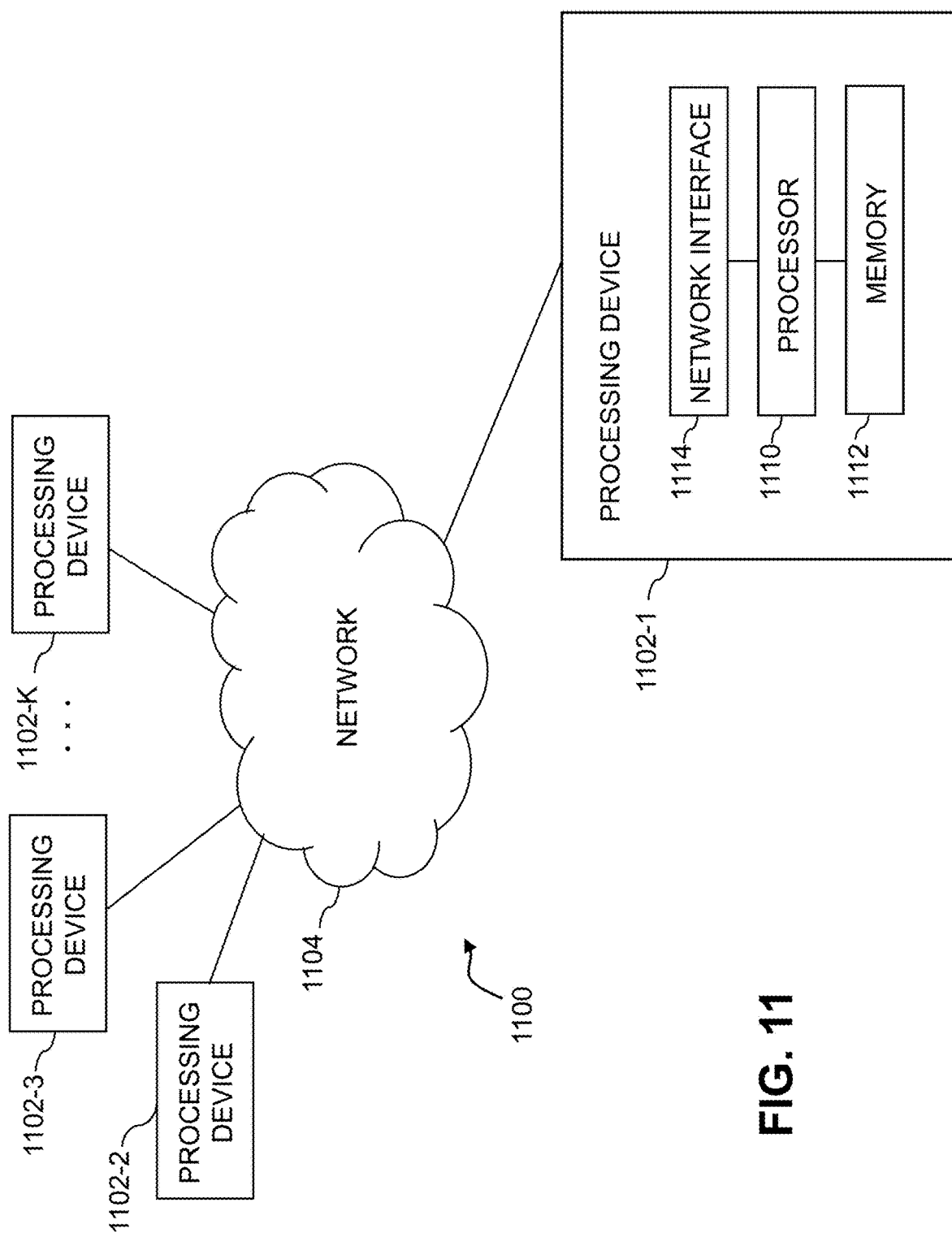


FIG. 11

DECENTRALIZED IDENTITY MANAGEMENT SYSTEM

FIELD

[0001] The field relates generally to information processing system and, more particularly, to decentralized identity management in such systems.

BACKGROUND

[0002] Identity management is a process of managing information about users on computing networks. The information typically includes information that authenticates the identity of a user to a computing network. In many scenarios, the identity of the user needs to be authenticated prior to a computing network giving the user access to one or more of data, a service, an application, or another computing resource controlled by the computing network.

[0003] Typically, identity management is achieved in a centralized manner meaning that every user in a given system environment typically has to communicate with the same central identity management authority. For example, each user that wishes to access some resource on a computing network must first prove his identity to a server controlled by a central authority. The central authority then either notifies the computing network that a given user is authenticated or provides the given user with an authentication token that the given user must provide to the computing network. In either case, every user must contact the central authority to obtain access to resources of any computing network. However, such centralized identity management presents many challenges.

SUMMARY

[0004] Illustrative embodiments of the invention provide decentralized identity management.

[0005] For example, in one illustrative embodiment, a method comprises the following steps. A given node associated with a plurality of nodes registers a decentralized identity for the given node on a decentralized identity blockchain. The registered decentralized identity is controlled by the given node and defined by an identity record stored on the decentralized identity blockchain. The registered decentralized identity for the given node is used to access one or more resources of a given decentralized application.

[0006] In one or more illustrative embodiments, the identity record comprises a human readable name of the given node, and a public key corresponding to a private key held by the given node. The private key is usable by the given node to prove that the given node controls the identity, while the public key stored in the identity is usable by at least a subset of the plurality of nodes to verify that the given node controls the identity. Further, the identity record comprises a decentralized file system address pointing to immutable data stored remote from the decentralized identity blockchain. The identity record further comprises a decentralized name space address pointing to mutable data stored remote from the decentralized identity blockchain.

[0007] In one or more illustrative embodiments, the registered decentralized identity for the given node is used to access the one or more resources of the given decentralized application by sending a request from the given node to the given decentralized application to enable the given decen-

tralized application to authenticate the given node by accessing the decentralized identity blockchain to obtain certificate-related data associated with the given node. An indication of authentication is received at the given node from the given decentralized application wherein access to the one or more resources of the given decentralized application is based on the indication of authentication, e.g., granted or denied.

[0008] Advantageously, illustrative embodiments overcome drawbacks associated with a centralized identity approach by providing decentralized identity management functionality that enables, by way of example only, elimination of a single point of system failure (e.g., a central identity server), a single identity for multiple resources or applications, user ownership of identity, balance between security and flexibility, improved scalability, improved availability, and/or reduction in latency.

[0009] These and other features and advantages of the invention will become more readily apparent from the accompanying drawings and the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 illustrates participants in a centralized identity management process.

[0011] FIG. 2 illustrates an example of authentication in a centralized identity management process.

[0012] FIG. 3 illustrates a decentralized identity management process according to an illustrative embodiment.

[0013] FIG. 4 illustrates use of file system and name space addresses in a decentralized identity management process according to an illustrative embodiment.

[0014] FIG. 5 illustrates interaction between an identity chain and a decentralized application in a decentralized identity management process according to an illustrative embodiment.

[0015] FIG. 6 illustrates interaction between an identity chain and an application chain in a decentralized identity management process according to an illustrative embodiment.

[0016] FIG. 7 illustrates an identity registration and permission request procedure in a decentralized identity management process according to an illustrative embodiment.

[0017] FIG. 8 illustrates an identity and permission verification procedure in a decentralized identity management process according to an illustrative embodiment.

[0018] FIG. 9 illustrates a decentralized identity management methodology according to an illustrative embodiment.

[0019] FIGS. 10 and 11 illustrate examples of processing platforms utilized to implement at least a portion of a decentralized identity management system according to illustrative embodiments.

DETAILED DESCRIPTION

[0020] Illustrative embodiments will be described herein with reference to exemplary information processing systems and associated host devices, storage devices and other processing devices. It is to be appreciated, however, that embodiments are not restricted to use with the particular illustrative system and device configurations shown. Accordingly, the term “information processing system” as used herein is intended to be broadly construed, so as to encompass, for example, processing systems comprising

cloud computing and storage systems, as well as other types of processing systems comprising various combinations of physical and virtual computing resources.

[0021] An information processing system may therefore comprise, for example, a cloud infrastructure hosting multiple tenants that share cloud computing resources. Such systems are considered examples of what are more generally referred to herein as cloud computing environments. Environments that implement multiple cloud platforms are referred to as multi-cloud computing environments. A multi-cloud computing environment employed by an enterprise may comprise two or more clouds (e.g., two or more private clouds, two or more public clouds, or a combination of one or more private clouds and one or more public clouds).

[0022] One or more cloud platforms that are part of this environment may be within the exclusive control and management of the enterprise, and therefore are considered “private clouds.” On the other hand, the cloud computing environment can include one or more cloud platforms that can be used by multiple enterprises, and not necessarily controlled or managed by any of the multiple enterprises but rather are controlled and managed by one or more third-party cloud providers. Such cloud platforms are typically considered “public clouds.” More typically, but not exclusively, enterprises may choose to host their data (as well as applications, workloads, services, etc.) on a combination of private and public clouds that form a multi-cloud computing environment.

[0023] One or more clouds may also be referred to as a “data center.” Alternatively, a data center may be comprised of one or more clouds. The term “enterprise” as used herein is intended to be broadly construed, and may comprise, for example, one or more businesses, one or more corporations or any other one or more entities, groups, or organizations. An “entity” as illustratively used herein may be a person or system.

[0024] As mentioned above, existing identity management is achieved in a centralized manner (identifiers management by a central authority). Illustrative embodiments realize that decentralized identity management using decentralized identifiers provides many advantages over centralized identifiers. With decentralized identifiers, the user can digitally identify himself/herself by holding a single private key. Moreover, only the user holding the private key, instead of one centralized server, owns the decentralized identifier.

[0025] Thus, as will be explained in detail herein, illustrative embodiments provide a decentralized identity management system. The system provides functionalities for the user and other entities to create, manage and use their own decentralized identifier. The proposed framework is also configured to interact with decentralized applications built on top of or otherwise in communication with this decentralized identity management system. One non-limiting example of a decentralized application that can be used with the decentralized identity management system is described in U.S. Ser. No. 16/048,785, filed on Jul. 30, 2018 and entitled “Decentralized Policy Publish and Query System for Multi-cloud Computing Environment,” the disclosure of which is incorporated by reference herein in its entirety. Some examples will be described below using this decentralized policy publication and query application. However, it is to be appreciated that the decentralized identity management system is not be limited in any way to the application in the decentralized policy publication and query

system described in U.S. Ser. No. 16/048,785. Prior to describing illustrative decentralized identity management embodiments, a brief illustrative explanation of centralized identity management will first be given.

[0026] FIG. 1 illustrates participants in a centralized identity management process **100**. As shown, a user **102** wants to request some resources **104** and authentication is required. The term “user” as used herein is understood to be an entity that interacts with the identity management system via a computer or other communication device (e.g., client device). “Resources,” as mentioned above, can include data, services, applications, combinations thereof, or any other computer-accessible resource. Assume a centralized server **106** holds resources **104** and also holds identities of users in an identity (ID) server **108**. The user **102** first contacts the ID server **108** to prove that he/she has the permission to retrieve the resources. However, the problem here is the user **102** does not own his/her identity. Instead, the centralized ID server **108** holds the identities. As such, it is possible that the identity is changed or even removed without notifying the user.

[0027] More recently, computing networks including ones that are public (e.g., Internet) and ones that are private (e.g., enterprise-controlled) have become more and more decentralized. For example, decentralized file systems and decentralized data protection systems have been proposed to manage and use data in a decentralized way. Nevertheless, in each of these decentralized resource solutions, the identity server is still centralized.

[0028] FIG. 2 illustrates an example of authentication in a centralized identity management process **200** in a policy file management system. More particularly, a central authority (CA) **202** provides a centralized identity management function. One or more administrators **204** oversee the CA operations. A client **206** associated with a given user **208** registers with CA **202**. Note that client **206** can be any client in the system. Once registered in the identity management system, client **206** can submit a request to one or more peers **210** which process the request using chaincode **212** and orderer(s) **214** in conjunction with a ledger **216**, e.g., a blockchain, as will be further described below.

[0029] In one implementation, the public key (of a public key/private key pair) of CA **202** is shared by all peers **210**, so that every peer on the chain can verify a certificate provided by client **206** (in its request) independently. More particularly, user **208** requests access to ledger **216** from CA **202**. CA **202** issues a certificate accordingly, where the certificate is digitally signed using the private key of CA **202**. User **208** tries to access (read/write) ledger **216** by providing the certificate issued by CA **202**. All peers **210** can verify the certificate independently and act accordingly. Note that orderer **214** (e.g., one orderer **214** per peer **210** in an illustrative implementation) is configured to keep track of policy file versions and how each peer receives them as needed. In some cases, different clients get different versions of the policy file at the same time.

[0030] Thus, data has already been stored on ledger **216**, which is part of a fully decentralized application. However, there is still one centralized certificate authority, i.e., CA **202**. The centralized CA introduces a single point of failure to the entire system, even though the data is stored on ledger/blockchain in a decentralized way.

[0031] Besides decentralization, illustrative embodiments realize that users can benefit from additional features in an identity management system including, by way of example:

[0032] (i) The user should own his identity and have full control over it. The identity should not be stored on a centralized server, where the user only has limited control.

[0033] (ii) The identity should have a human readable name. Decentralized identity such as a media access control (MAC) address is not controlled by any central authority, however, the MAC address is not human readable and thus cannot be easily used in daily work.

[0034] (iii) The user could use a single identity to manage several resources (possibly controlled by different authorities).

[0035] From the view of resource managers or application developers, they could also build the access control feature based on the decentralized identities.

[0036] In light of these realizations, a number of challenges arise.

[0037] Single Point of Failure.

[0038] As the Internet becomes more and more decentralized, the user may have cryptocurrency stored on distributed ledgers and have valuable data stored on decentralized file systems, such as the Interplanetary File System (IPFS). However, the centralized identity server acts as the bottleneck, introducing a single point of failure to the otherwise fully decentralized solution.

[0039] Single Identity for Multiple Resources.

[0040] The resource owners typically run several different centralized servers of identities for their own purposes. If one user wants to have access to different resources, he/she should register multiple identities. This introduces a heavy burden to the user for managing different identities for different resources.

[0041] Ownership of Identity.

[0042] In the centralized solution, the identity registered by the user on the centralized server is not actually owned by the user. The data is stored on the server, so that the server owner actually has the power to modify and even delete the identity.

[0043] Flexibility Versus Security.

[0044] When it comes to a solution based on the blockchain, there is a tradeoff between flexibility and security. Due to the immutability of the blockchain, the entire identity record could be stored on the blockchain. This provides maximum security. However, updating the identity can not be easily achieved in a cost efficient way. On the other hand, if the identity is stored on some mutable storage, the integrity of the identity cannot easily be guaranteed.

[0045] Scalability.

[0046] If the entire set of identity records are stored on the blockchain, there is too much data directly stored on the chain. Since every node of the blockchain should maintain a full copy of the data on the chain, this introduces a scalability problem. The size of the chain could grow very fast, making it non-feasible in real world applications. Moreover, any update to the identity record will result in a new transaction as well as a new block (or a new record on the new block) on the chain, introducing significant computation burden on the entire network.

[0047] Availability.

[0048] The identity once created should have high availability. If the centralized server is used, the availability of the

identity cannot be guaranteed, since the centralized server introduces a single point of failure.

[0049] Latency.

[0050] If all the identities are stored on the chain, high latency is experienced, since all users will contact the peers of the blockchain to retrieve the identities, and the bandwidth of peers becomes the bottleneck.

[0051] Illustrative embodiments overcome the above and other drawbacks associated with a centralized approach to identity management. In one or more illustrative embodiments, a combination of blockchain and IPFS are used to address the above-highlighted and other problems. While IPFS is described herein according to one or more illustrative embodiments, it is to be understood that alternative embodiments are not limited to IPFS.

[0052] IPFS is an open-source protocol that provides a decentralized method of storing and sharing files relying on a content-addressable, peer-to-peer hypermedia distribution. The compute nodes in an IPFS network form a distributed file system. A given one of the IPFS compute nodes resides on a client (or client otherwise has access to an IPFS compute node). The IPFS protocol was developed to replace the HyperText Transfer Protocol (HTTP) of the Internet which relies on location addressing (i.e., using Internet Protocol (IP) addresses to identify the specific computing resource that is hosting a desired data set). As such, with HTTP, the subject data set must be retrieved from the computing resource where it originated or some computing resource within the content delivery network (CDN) each time the data set is requested.

[0053] IPFS operates by operatively coupling devices/nodes with the same system of files. More particularly, IPFS uses a distributed hash table (DHT) with a block exchange (BitSwap) and namespace methodology that operates across disparate devices and unreliable networks. IPFS operates similarly to a torrent system, except that rather than exchanging media, IPFS exchanges objects based on a key-value data store. Any type of content can be inserted into the data sharing system, and the system returns a key (i.e., in the form of a hash value) that can be used to retrieve the content from a node that has it stored thereon at any time. Accordingly, IPFS is a content addressing protocol instead of a location addressing protocol. That is, the hash value is independent of the origin of the data set and can be hosted anywhere in the system.

[0054] In one example, the IPFS system is further described in J. Benet, "IPFS—Content Addressed, Versioned, P2P File System," 2014, the disclosure of which is incorporated by reference herein in its entirety. However, illustrative embodiments are not limited to this particular data storage network and alternative systems may be employed.

[0055] Furthermore, as used herein, the terms "blockchain," "chain," "ledger" and "blockchain ledger" may be used interchangeably. As is known, the blockchain or distributed ledger protocol is implemented via a distributed, decentralized computer network of compute nodes. A given one of the blockchain compute nodes resides on a client (or client otherwise has access to a blockchain compute node). The compute nodes are operatively coupled in a peer-to-peer communications protocol. In the computer network, each compute node is configured to maintain a blockchain which is a cryptographically secured record or ledger of data blocks that represent respective transactions within a given

computational environment. The blockchain is secured through use of a cryptographic hash function. A cryptographic hash function is a cryptographic function which takes an input (or “message”) and returns a fixed-size alphanumeric string, which is called the hash value (also a message digest, a digital fingerprint, a digest, or a checksum). Each blockchain is thus a growing list of data records hardened against tampering and revision, and typically includes a timestamp, current transaction data, and information linking it to a previous block. More particularly, each subsequent block in the blockchain is a data block that includes a given transaction(s) and a hash value of the previous block in the chain, i.e., the previous transaction. That is, each block is typically a group of transactions. Thus, advantageously, each data block in the blockchain represents a given set of transaction data plus a set of all previous transaction data. In digital ledger technologies such as blockchain, an underlying consensus algorithm is typically used to validate new transactions before they are added to the distributed ledger. Typically, for example, the new transaction is broadcast to all nodes within the network, inspected, a consensus is reached, and the entry is formally committed to the blockchain based on whether consensus is reached that the entry is validated.

[0056] In the case of a “bitcoin” type implementation of a blockchain distributed ledger, the blockchain contains a record of all previous transactions that have occurred in the bitcoin network. The bitcoin system was first described in S. Nakamoto, “Bitcoin: A Peer to Peer Electronic Cash System,” 2008, the disclosure of which is incorporated by reference herein in its entirety.

[0057] A key principle of the blockchain is that it is trusted. That is, it is critical to know that data in the blockchain has not been tampered with by any of the compute nodes in the computer network (or any other node or party). For this reason, a cryptographic hash function is used. While such a hash function is relatively easy to compute for a large data set, each resulting hash value is unique such that if one item of data in the blockchain is altered, the hash value changes. However, it is realized that given the constant generation of new transactions and the need for large scale computation of hash values to add the new transactions to the blockchain, the blockchain protocol rewards compute nodes that provide the computational service of calculating a new hash value. In the case of a bitcoin network, a predetermined number of bitcoins are awarded for a predetermined amount of computation. The compute nodes thus compete for bitcoins by performing computations to generate a hash value that satisfies the blockchain protocol. Such compute nodes are referred to as “miners.” Performance of the computation of a hash value that satisfies the blockchain protocol is called “proof of work.” While bitcoins are one type of reward, blockchain protocols can award other measures of value (monetary or otherwise) to successful miners.

[0058] It is to be appreciated that the above description represents an illustrative implementation of the blockchain protocol and that embodiments are not limited to the above or any particular blockchain protocol implementation. As such, other appropriate processes may be used to securely maintain and add to a set of data in accordance with embodiments of the invention. For example, distributed ledgers such as, but not limited to, R3 Corda, Ethereum, and Hyperledger may be employed in alternative embodiments.

[0059] FIG. 3 illustrates an example 300 of decentralized identity management according to an illustrative embodiment. As shown, an identity chain (blockchain) 310 is used to store identity records 312 through 318. Since the blockchain is decentralized, the identities thus are stored in a decentralized manner. In addition, in this illustrative embodiment, all parts of the identity are not stored on the blockchain 310. Rather, in one or more illustrative embodiments, only the addresses (or pointers) are stored on the blockchain 310, and the actual data is stored off (remote from) the blockchain on a decentralized file system (such as IPFS).

[0060] A definition of the decentralized identity is illustrated in FIG. 3. Details for records 312, 316 and 318 are shown as 313, 317 and 319, respectively. As shown, each record is key-value based. The key of the record is a human readable name, for example, user.id and CA1.id, instead of 01:23:45:67:89:ab. The value of the record is a triple.

[0061] The first entry of the triple is the public key (Pk) corresponding to the private key held by the owner of the identity. The user can use the private key to prove that he actually controls the identity, while anyone can use the public key stored in the identity to verify that claim.

[0062] The second entry is the IPFS address, pointing to some immutable data. Note that the identity stored on the blockchain and the data stored on the IPFS are both immutable. From the IPFS address in the identity record, the immutable data held by the user is retrieved. Due to the immutability, rarely changed data is preferably stored in this IPFS address.

[0063] The third entry of the triple is an Interplanetary Name Space (IPNS) address. IPNS is a namespace on top of IPFS, see, e.g., the above-referenced J. Benet, “IPFS—Content Addressed, Versioned, P2P File System,” 2014. IPNS provides mutability to store and retrieve data. IPNS can be treated as a folder, where only the one holding the private key can store data in this folder, while no one else can store data into this folder but they can read data from the folder. The frequently changed data can be stored in the location (remote from the identity blockchain) that is pointed to by the IPNS address. In a decentralized authentication case, certificates 320 of the user to prove the user has certain permission can be stored via the IPNS address. Likewise, revoked certificates 322 held by system administrators can be stored via the IPNS address. This illustrative record structure is shown in further detail in FIG. 4.

[0064] More particularly, FIG. 4 illustrates use of file system and name space addresses in decentralized identity management according to an illustrative embodiment. In example 400, blockchain 410 includes a record 412 which includes an IPFS address that stores (i.e., points to the storage of) traceable information 414 whereby changes to the information result in a new transaction (tx) on the blockchain. Client information (e.g., client device ID) can be stored in this entry of the record. Record 412 also includes an IPNS address that stores (i.e., point to storage of) mutable information 416 whereby changes can more easily be made as explained above. Certificates and revoked certificates can be stored in this entry of the record.

[0065] With identities managed in a decentralized manner according to illustrative embodiments, an improved solution to the authentication problem illustrated in FIG. 2 is achieved. FIG. 5 illustrates an example 500 of interaction between an identity chain 510 and a decentralized applica-

tion chain (policy chain) **520** according to an illustrative embodiment. Note again that the term chain as used herein is short for blockchain.

[0066] Both the user and the administrator of the application register themselves by creating an identity record on the identity chain **510**. The identity records are created as described above in the context of FIGS. **3** and **4**. In this example **500** in FIG. **5**, assume record **512** is the user identity record and record **514** is the administrator identity record. While in the policy (application) chain **520**, the authentication procedure is implemented by only referring to the human readable ID names on the identity chain, i.e., record **522** refers to the user ID and record **524** refers to the administrator ID.

[0067] For example, in a decentralized policy publication and query system (e.g., as described in the above-referenced U.S. Ser. No. 16/048,785), the policy chain used to publish a policy file also stores the ID name of the administrator of the policy chain **520**. Anyone can publish and query a policy file on the policy chain **520** by providing his or her ID name. The policy chain **520** automatically checks both the user identity and the administrator identity from the identity chain **510** to decide whether the user has certain permission. **[0068]** Note that in this way, the requirement of using centralized authentication or an identity server is eliminated, so that the service (e.g., policy publication and query) can be provided in a fully decentralized way.

[0069] FIG. **6** illustrates an overall process **600** of the interaction between an identity chain **610** and an application (policy) chain **620**. Steps 0 through 5 in process **600** will be explained in detail below in the context of FIGS. **7** and **8**.

[0070] In general, in FIG. **6**, the user and the administrator register themselves on the identity chain **610** in step 0. The user then submits an access request to the policy chain **620** in step 1. In step 2, the policy chain **620** contacts the identity chain **610** to authenticate the user. The identity chain **610** retrieves authentication data (e.g., certificate) using the user identity in step 3, and responds to the policy chain **620** in step 4. The policy chain **620**, in step 5, then responds to the user giving the user access consistent with the permission(s) in the certificate or otherwise denying access.

[0071] The registration process **700** is illustrated in FIG. **7** in the context of an administrator (admin) **702**, a user **704**, an identity chain (channel) **710** and a policy chain (channel) **720**, and steps a through e. Note that the sequential order of letters a through e in FIG. **7** do not necessarily reflect the order in which the corresponding steps are explained in the description below. Note that the terms chain and channel are interchangeable in this illustrative description.

[0072] More particularly, as shown in FIG. **7**, user **704** prepares the user name (e.g. 'user.id'), the public and private key pair, and the IPNS/IPFS addresses to register himself (step c).

[0073] The data to be written in the identity chain **710** thus includes {user.id, public key, IPFS/IPNS addresses}, and the user digitally signs the data with the private key:

[0074] Signature=PriKey(Hash({user.id, public key, IPFS/IPNS addresses}))

[0075] The peers of identity chain **710** receive the registration request:

[0076] (i) Check whether or not the user.id has been used.

[0077] (ii) Verify that the user does have the private key:

[0078] PubKey(Signature) Hash({user.id, public key, IPFS/IPNS address})

[0079] (iii) If the request passes the verification, accept it and commit the transaction.

[0080] The user **704** requests certain permission from the administrator **702** which is also illustrated in process **700** of FIG. **7** (step d).

[0081] It is assumed that there is an admin ID of the policy chain **720**, which defines who has read/write access to the policy chain **720**. The admin ID ('CA1.id' in this example) is registered on the identity chain **720** (step a), and written into the genesis block of the policy chain **720** (step b) since the policy chain is set up by the administrator.

[0082] The administrator controls his/her own private key. The revoked list contains a set of users whose access to the policy chain **720** has been revoked by the administrator.

[0083] A user with id 'user.id' can freely register himself/herself on the identity chain **710**, however, this id has no access to the policy chain **720**. The user can request access from the administrator, and the administrator may issue a certificate to the user accordingly. In one or more illustrative embodiments, the certificate has {expire date, user.id, permissions granted (read or write or both), issuer (id name and public key)} and is signed by the private key of the administrator (where the corresponding public key is stored in the identity record of the administrator on the identity chain **710**).

[0084] It is to be appreciated that the process is done off-chain, however, it is also possible to operate on chain, at the cost of significantly more transactions.

[0085] Once the user receives the certificate, the user saves it in his/her own [IPNS address] recorded on the identity chain **710** (step e).

[0086] The administrator can revoke the certificates issued, and stores the revoked certificates in his/her own [IPNS address].

[0087] Next, the user publishes a new policy file to the policy chain **720**. The process **800** is illustrated in FIG. **8** in the context of steps 1 through 5. Note that policy publication is only one example of a decentralized application with which decentralized identity management embodiments can be implemented.

[0088] The user prepares a new request, which contains the new policy file and his own id ('user.id'). The request (with nonce) is signed by the user's private key. The user submits the request to the policy chain **720** (step 1).

[0089] Peers on the policy chain **720** who receive the request:

[0090] (i) Retrieve the public key and revoked list of 'CA1.id' from the identity chain **710** (part of step 2). The information is retrievable since the admin id 'CA1.id' is saved on the policy chain **720**. This procedure can be optimized. For example, once the public key is retrieved, it can be saved, and all peers can auto-synchronize with the revoked list.

[0091] (ii) Retrieve the public key and corresponding certificate of the user, based on the 'user.id' in the request (part of step 2).

[0092] (iii) Verify the request using the user's public key, making sure that the request is actually submitted by 'user.id' (step 3).

[0093] (iv) Verify the certificate of the user using the public key of 'CA1.id', making sure that the certificate is not faked (step 4).

[0094] (v) Check the certificate for corresponding permissions to operate (step 5).

[0095] If all verifications are satisfied, the peers commit the transaction and publish the new policy along with the 'user.id'. The policy and 'user.id' are signed by the user's private key, so that the entire network can repeat the verification process independently later.

[0096] Given the illustrative description of decentralized identity management techniques herein, FIG. 9 depicts a methodology 900 for decentralized identity management according to an illustrative embodiment.

[0097] In step 902 a given node associated with a plurality of nodes registers a decentralized identity for the given node on a decentralized identity blockchain. The registered decentralized identity is controlled by the given node and defined by an identity record stored on the decentralized identity blockchain.

[0098] In step 904, the registered decentralized identity for the given node is used to access one or more resources of a given decentralized application.

[0099] Decentralized identity management according to illustrative embodiments has many advantages, examples of which will now be described.

[0100] No Network Bottleneck.

[0101] With only the addresses of data stored on the blockchain and the actual data stored off chain, the network bandwidth limitation of blockchain peers will not be the bottleneck. Moreover, in illustrative embodiments that employ IPFS, which is peer-to-peer based, the bandwidth limitation of a single peer will not be the bottleneck.

[0102] Small Data on Chain with High Availability.

[0103] In the decentralized identity management system described herein, only data addresses are stored on the blockchain and, as such, the size of the blockchain will grow slowly. Furthermore, since both addresses to mutable contents and immutable contents are provided, the user could store constantly changing data in the IPNS address, limiting the number of new transactions and blocks. Moreover, data stored on the IPFS can be retrieved using the content-based address (CA) regardless of where and how many copies are stored, thus providing high availability of the data.

[0104] High Security and Flexibility.

[0105] Security and flexibility are both achieved by the decentralized identity management system described herein. Security is guaranteed by the immutability of both the blockchain and IPFS. Flexibility is provided by the IPNS namespace, since mutable data is stored in accordance with the identity record, and the integrity/security of the data stored in IPNS is guaranteed by the public key infrastructure (PKI) based procedure.

[0106] One Identity for Multiple Applications.

[0107] In the decentralized identity management system described herein, only the user holding the private key corresponding to the public key stored in the record owns the record. The user can store as many certificates as he wants in the IPNS address for different applications. So once a user owns the identity, he can use it for multiple applications.

[0108] It is to be appreciated that the particular advantages described above and elsewhere herein are associated with particular illustrative embodiments and need not be present in other embodiments. Also, the particular types of information processing system features and functionality as illustrated in the drawings and described above are exemplary only, and numerous other arrangements may be used in other embodiments.

[0109] Illustrative embodiments of processing platforms utilized to implement host devices and storage systems with decentralized identity management functionality will now be described in greater detail with reference to FIGS. 10 and 11.

[0110] FIG. 10 shows an example processing platform comprising cloud infrastructure 1000. The cloud infrastructure 1000 comprises a combination of physical and virtual processing resources that may be utilized to implement at least a portion of the decentralized identity management system described herein. The cloud infrastructure 1000 comprises multiple virtual machines (VMs) and/or container sets 1002-1, 1002-2, . . . 1002-L implemented using virtualization infrastructure 1004. The virtualization infrastructure 1004 runs on physical infrastructure 1005, and illustratively comprises one or more hypervisors and/or operating system level virtualization infrastructure. The operating system level virtualization infrastructure illustratively comprises kernel control groups of a Linux operating system or other type of operating system.

[0111] The cloud infrastructure 1000 further comprises sets of applications 1010-1, 1010-2, . . . 1010-L running on respective ones of the VMs/container sets 1002-1, 1002-2, . . . 1002-L under the control of the virtualization infrastructure 1004. The VMs/container sets 1002 may comprise respective VMs, respective sets of one or more containers, or respective sets of one or more containers running in VMs.

[0112] In some implementations of the FIG. 10 embodiment, the VMs/container sets 1002 comprise respective VMs implemented using virtualization infrastructure 1004 that comprises at least one hypervisor. Such implementations can provide functionality of the type described above for one or more processes running on a given one of the VMs. For example, each of the VMs can implement functionality for one or more processes running on that particular VM.

[0113] An example of a hypervisor platform that may be used to implement a hypervisor within the virtualization infrastructure 1004 is the VMware® vSphere® which may have an associated virtual infrastructure management system such as the VMware® vCenter™. The underlying physical machines may comprise one or more distributed processing platforms that include one or more storage systems.

[0114] In other implementations of the FIG. 10 embodiment, the VMs/container sets 1002 comprise respective containers implemented using virtualization infrastructure 1004 that provides operating system level virtualization functionality, such as support for Docker containers running on bare metal hosts, or Docker containers running on VMs. The containers are illustratively implemented using respective kernel control groups of the operating system. Such implementations can provide decentralized identity management functionality of the type described above for one or more processes running on different ones of the containers. For example, a container host device supporting multiple containers of one or more container sets can implement one or more instances of decentralized identity management logic for use in implementing the functionalities described herein.

[0115] As is apparent from the above, one or more of the processing modules or other components of system 100 may each run on a computer, server, storage device or other processing platform element. A given such element may be

viewed as an example of what is more generally referred to herein as a “processing device.” The cloud infrastructure **1000** shown in FIG. **10** may represent at least a portion of one processing platform. Another example of such a processing platform is processing platform **1100** shown in FIG. **11**.

[0116] The processing platform **1100** in this embodiment comprises a portion of system **100** and includes a plurality of processing devices, denoted **1102-1**, **1102-2**, **1102-3**, . . . **1102-K**, which communicate with one another over a network **1104**.

[0117] The network **1104** may comprise any type of network, including by way of example a global computer network such as the Internet, a WAN, a LAN, a satellite network, a telephone or cable network, a cellular network, a wireless network such as a WiFi or WiMAX network, or various portions or combinations of these and other types of networks.

[0118] The processing device **1102-1** in the processing platform **1100** comprises a processor **1110** coupled to a memory **1112**.

[0119] The processor **1110** may comprise a microprocessor, a microcontroller, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other type of processing circuitry, as well as portions or combinations of such circuitry elements.

[0120] The memory **1112** may comprise random access memory (RAM), read-only memory (ROM), flash memory or other types of memory, in any combination. The memory **1112** and other memories disclosed herein should be viewed as illustrative examples of what are more generally referred to as “processor-readable storage media” storing executable program code of one or more software programs.

[0121] Articles of manufacture comprising such processor-readable storage media are considered illustrative embodiments. A given such article of manufacture may comprise, for example, a storage array, a storage disk or an integrated circuit containing RAM, ROM, flash memory or other electronic memory, or any of a wide variety of other types of computer program products. The term “article of manufacture” as used herein should be understood to exclude transitory, propagating signals. Numerous other types of computer program products comprising processor-readable storage media can be used.

[0122] Also included in the processing device **1102-1** is network interface circuitry **1114**, which is used to interface the processing device with the network **1104** and other system components, and may comprise conventional transceivers.

[0123] The other processing devices **1102** of the processing platform **1100** are assumed to be configured in a manner similar to that shown for processing device **1102-1** in the figure.

[0124] Again, the particular processing platform **1100** shown in the figure is presented by way of example only, and system **100** may include additional or alternative processing platforms, as well as numerous distinct processing platforms in any combination, with each such platform comprising one or more computers, servers, storage devices or other processing devices.

[0125] For example, other processing platforms used to implement illustrative embodiments can comprise converged infrastructure such as VxRail™, VxRack™, VxRack™ FLEX, VxBlock™ or Vblock® converged infra-

structure from VCE, the Virtual Computing Environment Company, now the Converged Platform and Solutions Division of Dell EMC.

[0126] It should therefore be understood that in other embodiments different arrangements of additional or alternative elements may be used. At least a subset of these elements may be collectively implemented on a common processing platform, or each such element may be implemented on a separate processing platform.

[0127] As indicated previously, components of an information processing system as disclosed herein can be implemented at least in part in the form of one or more software programs stored in memory and executed by a processor of a processing device. For example, at least portions of the decentralized identity management functionality of one or more components of a host device or storage system as disclosed herein are illustratively implemented in the form of software running on one or more processing devices.

[0128] It should again be emphasized that the above-described embodiments are presented for purposes of illustration only. Many variations and other alternative embodiments may be used. For example, the disclosed techniques are applicable to a wide variety of other types of information processing systems, host devices, storage systems, storage nodes, storage devices, storage controllers, decentralized identity management processes and associated control logic. Also, the particular configurations of system and device elements and associated processing operations illustratively shown in the drawings can be varied in other embodiments. Moreover, the various assumptions made above in the course of describing the illustrative embodiments should also be viewed as exemplary rather than as requirements or limitations of the disclosure. Numerous other alternative embodiments within the scope of the appended claims will be readily apparent to those skilled in the art.

What is claimed is:

1. A method comprising:

registering, by a given node associated with a plurality of nodes, a decentralized identity for the given node on a decentralized identity blockchain, wherein the registered decentralized identity is controlled by the given node and defined by an identity record stored on the decentralized identity blockchain; and

using the registered decentralized identity, for the given node, to access one or more resources of a given decentralized application;

wherein the given node is implemented via at least one processing device comprising a processor coupled to a memory.

2. The method of claim 1, wherein the identity record comprises a human readable name of the given node.

3. The method of claim 1, wherein the identity record comprises a public key corresponding to a private key held by the given node.

4. The method of claim 3, wherein the private key is usable by the given node to prove that the given node controls the identity.

5. The method of claim 4, wherein the public key stored in the identity is usable by at least a subset of the plurality of nodes to verify that the given node controls the identity.

6. The method of claim 1, wherein the identity record comprises a decentralized file system address pointing to immutable data stored remote from the decentralized identity blockchain.

7. The method of claim 6, wherein the immutable data comprises data associated with the given node that infrequently or never changes.

8. The method of claim 1, wherein the identity record comprises a decentralized name space address pointing to mutable data stored remote from the decentralized identity blockchain.

9. The method of claim 8, wherein the mutable data comprises certificate-related data associated with the given node.

10. The method of claim 1, wherein the identity record comprises information associated with the given node in a key-value pair format.

11. The method of claim 1, wherein the step of using the registered decentralized identity, for the given node, to access one or more resources of a given decentralized application further comprises:

sending a request from the given node to the given decentralized application to enable the given decentralized application to authenticate the given node by accessing the decentralized identity blockchain to obtain certificate-related data associated with the given node; and

receiving an indication of authentication at the given node from the given decentralized application wherein access to the one or more resources of the given decentralized application is based on the indication of authentication.

12. The method of claim 11, wherein the given decentralized application is maintained in accordance with a decentralized application blockchain in communication with the decentralized identity blockchain.

13. An article of manufacture comprising a non-transitory processor-readable storage medium having stored therein program code of one or more software programs, wherein the program code when executed by at least one processing device causes the processing device to perform steps of:

registering, by a given node associated with a plurality of nodes, a decentralized identity for the given node on a decentralized identity blockchain, wherein the regis-

tered decentralized identity is controlled by the given node and defined by an identity record stored on the decentralized identity blockchain; and

using the registered decentralized identity, for the given node, to access one or more resources of a given decentralized application

14. An apparatus comprising at least one processing device, wherein the at least one processing device comprises a processor coupled to a memory configured to:

register a decentralized identity, by a given node associated with a plurality of nodes, on a decentralized identity blockchain, wherein the registered decentralized identity is controlled by the given node and defined by an identity record stored on the decentralized identity blockchain; and

use the registered decentralized identity, for the given node, to access one or more resources of a given decentralized application.

15. The apparatus of claim 14, wherein the identity record comprises a human readable name of the given node.

16. The apparatus of claim 14, wherein the identity record comprises a public key corresponding to a private key held by the given node.

17. The apparatus of claim 16, wherein the private key is usable by the given node to prove that the given node controls the identity.

18. The apparatus of claim 17, wherein the public key stored in the identity is usable by at least a subset of the plurality of nodes to verify that the given node controls the identity.

19. The apparatus of claim 14, wherein the identity record comprises a decentralized file system address pointing to immutable data stored remote from the decentralized identity blockchain.

20. The apparatus of claim 14, wherein the identity record comprises a decentralized name space address pointing to mutable data stored remote from the decentralized identity blockchain.

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